

The Trust Process Applied to Machine Learning Algorithms: The Influence of Calibrated Confidence Estimates

Gene M. Alarcon , Sarah A. Jessup , Scott K. Meyers , Sasha M. Willis, Krista N. Harris , Dexter Johnson, Jacob Noblick, and Walter D. Bennette

Abstract—Little research has been conducted on user interactions with machine learning models and users' trust. The current study explores differences in psychological perceptions of trust between two machine learning models. Across three studies, participants ($N = 130, 134, 132$) were recruited online and completed an image-binning monitoring task with either an uncalibrated classification model (UCM) or a calibrated classification model (CCM) that provided confidence estimates about their decisions. The UCM was highly confident, regardless of images from out-of-distribution classes, whereas the CCM was better calibrated between accuracy and confidence of out-of-distribution images. Results revealed participants performed better on the task in the CCM condition across all studies. In addition, trustworthiness perception results differed depending on whether the study included out-of-distribution images. Overall, calibrated confidence estimates displayed by CCMs can be an effective means of increasing transparency and enhancing our understanding of trust in machines, but more research is needed on trust perceptions in image classification tasks.

Index Terms—Human-machine interaction, machine learning (ML), reliance, transparency, trust.

I. INTRODUCTION

MACHINE learning (ML) and artificial intelligence (AI) technologies are quickly changing the landscape of modern work as they are used in a variety of contexts [1]. Often, the most accurate models are from a class of models known as deep neural networks (DNNs) [2]. Although these models

are typically the most accurate, they are often opaque, and the underlying metrics used to render a decision are unknown to the user, which can undermine trust [3]. In the past decade, researchers have attempted to develop several models that offer a deeper understanding of the underlying processes the DNNs use to make their decisions, including but not limited to explainable AI (XAI) and calibrated classification models (CCMs) [4]. Traditional models, also known as uncalibrated classification models (UCMs), typically have high confidence estimates, even if the stimuli are from classes outside the labeled training dataset (i.e., out-of-distribution data) [5]. CCMs have a built-in loss function that reduces confidence when aspects of the data do not fit the trained model classification categories, which leads to more accurate calibration of the confidence estimates. Although these models have become usable algorithms in a computer scientist's toolbox to enhance system transparency, there is little research that has explored the effects on the end-users' perceptions of the models and whether CCMs improve task performance. The current study explored differences in psychological perceptions of trust between traditional UCMs and newly developed CCMs.

A. Human-Machine Trust Process

Trust is essential for the successful adoption and utilization of AI systems [3]. Trust is the willingness to rely on another without the ability to monitor the referent [6]. Trust in machines can be delineated into four different components: individual differences, trustworthiness perceptions, reliance intentions (RI), and actual reliance behaviors [7]. In this article, we focus on trustworthiness perceptions, RI, and reliance behaviors. Trustworthiness perceptions are judgments of the system by the user and are comprised of purpose, process, and performance [7]. Purpose perceptions are the beliefs about whether the system is being used properly and in accordance with its design. Process perceptions describe how well the user believes they understand how the machine completes the task. Performance perceptions refer to the user's understanding of the machine's ability to perform the task. RI are the attitudes a user has toward system use and a willingness to be vulnerable [7], [8]. Trustworthiness will be different across systems [7] because different types of ML and AI models display different types and levels of transparency [3]. These differences in transparency can lead

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to different perceptions of purpose, process, and performance across systems, as well as subsequent RI and behaviors.

B. Machine Learning and Artificial Intelligence

ML/AI models are quickly becoming the dominant form of technology in many areas of human-machine teaming [1], [3], including image classification tasks [9]. The output of image classification models is typically a categorization of an image into one of at least two categories with confidence estimates (also known as uncertainty estimates), indicating the predicted probability of the classification being correct [10]. However, traditional ML/AI image-binning models are often overconfident when classifying images that represent classes for which no training data were provided (i.e., “out-of-distribution” images) [4]. This could lead to user miscalibration of their trust in the system, over-relying on the system when it does not alert them to potential errors or under-relying on the system by monitoring all of its decisions. CCMs, in contrast, are a new development in the field of computer science, which are sensitive to out-of-distribution data (i.e., data outside of the possible classification categories; see [4] for a full discussion of CCMs). As such, CCMs will still incorrectly classify out-of-distribution images, but will do so with lower confidence, which, when displayed appropriately, should alert the user to issues in the dataset or outcome decision.

C. Trust and AI/ML

Trust has been demonstrated as a key issue in the application of AI/ML models over the past decade. Indeed, both researchers and governments have called for more transparent AI/ML algorithms to facilitate trust and proper reliance on the models [1], [3]. Although there have been many new model methods developed to help facilitate understanding, the influence of this transparency on human perceptions and performance remains unclear, as less than 1% of research has explored human-AI/ML interactions from an experimental standpoint [11].

D. Confidence Estimates

AI/ML models operate on each prediction as a probability of classification based on the data provided. Research has demonstrated that adding the classification scores for each prediction can facilitate proper trust calibration, as the user is able to better understand when the model is less certain about a prediction [12]. Zhang et al. [12] demonstrated the utility of classification scores when the task was predicting nonimage criteria, such as salary. Research has also demonstrated that confidence estimates can reduce overreliance on AI/ML [13]. However, the authors are not aware of research that has explored the confidence estimates of AI/ML models in image classification tasks. AI/ML image classification models are increasingly being used in areas such as breast cancer screening [14]. Users may trust image classification models differently. Humans are often poor at interpreting numeric information, which may explain some of the results of the previous research [12], as they were primarily numeric classification tasks. Image classification is more complex for

AI/ML than humans as it involves real-world complexity and data limitations, and machines lack common sense [15]. As such, the effects of confidence scores on image classification have yet to be explored in human-AI/ML interactions.

II. CURRENT STUDY

The current research sought to explore differences between UCMs and CCMs on task performance, trustworthiness perceptions, and RI in an experimental task across three studies. As noted above, CCMs provide more accurate confidence estimates, as their parameters allow for decreased confidence in the classification of images that are outside the training dataset. Previous research [12] has demonstrated that confidence estimates are important trust indicators for users, but the previous tasks were forecasting tasks, whereas CCMs are focused on image classification tasks, with machines excelling in the former and humans excelling in the latter. Image classification tasks can have out-of-distribution data, whereas forecasting tasks do not. The more accurate confidence estimates from CCMs should lead to increased task performance for the user, as the model more accurately alerts the user to issues with image classification confidence, notably when images fall outside the training dataset. In contrast, traditional UCMs typically have high confidence in their image classifications, even when the image does not fit the criteria for the binning task the model was trained on. As such, we hypothesize the following.

Hypothesis 1: Participants will perform significantly better when monitoring the CCM than the traditional UCM.

Hypothesis 2: Decision time will be significantly lower when monitoring the CCM than the traditional UCM.

As noted earlier, much of the literature on transparency in AI/ML to date has focused on aspects that make models more explainable [1], [3], [16]. However, less than 1% of research has explored user perceptions of AI/ML models [11]. It remains to be seen if CCMs influence trustworthiness perceptions differently than UCMs. The increased accuracy of the confidence estimates for CCMs provides additional information about the performance of the model in the given context than traditional UCMs. We theorize that participants teamed with CCMs will have increased performance perceptions, as the calibrated confidence estimates are most closely associated with the performance construct of trustworthiness and provide more accurate information about the system’s performance than UCMs.

Hypothesis 3: Participants monitoring the CCM will have higher performance perceptions than those monitoring the UCM.

Solberg et al. [17] noted that manipulating aspects of interpersonal trust leads to changes in all aspects of trustworthiness because the facets of trustworthiness are all strongly correlated, and the dynamic between performance, purpose, and process may be different. As such, we were interested in exploring these dynamics. We may also see CCMs influencing perceptions of purpose and process. It is yet to be determined if CCMs solely impact performance perceptions. For example, the lack of confidence indicated by the CCM when classifying images outside its training dataset could indicate the improper use of the system,

which refers to the purpose aspect of trustworthiness. Similarly, the more accurate confidence estimates could relate information that the user can infer about the underlying decision-making process of the system (i.e., process information). As such, we test the postulate that CCMs also affect purpose and process perceptions.

Exploratory Question 1: Does monitoring the CCM influence purpose and process perceptions differently than monitoring the UCM?

This article conducted three studies to determine the differences in task performance and trustworthiness perceptions between CCMs and traditional UCMs. In all studies, participants were paired with an image-binning algorithm that was based on either a traditional UCM or a CCM. In Study 1, we explored the influence of confidence estimates in UCMs and CCMs with no out-of-distribution data on task performance, trustworthiness perceptions, and RI. We explored these differences because it is unclear how confidence estimates influence the task overall, as image classification is different than the previous forecasting tasks with confidence estimates [12]. In traditional human-machine teaming research, such as forecasting, there is a clear correct or incorrect response from the machine [12]. Out-of-distribution data add complexity to this relationship as output can be correct but low confidence or incorrect and low confidence, which may influence trust perceptions. In Study 2, we explored the influence of UCMs and CCMs in a dataset with out-of-distribution data on the same criteria to determine the differences, if any, in the criteria. This study explored the differences in a task consistent with the development of CCMs. Finally, in Study 3, we attempted to replicate Study 2 but also explored differences in qualitative responses to UCMs and CCMs to gain additional insight into our quantitative findings. This article is one of the first to explore human-AI/ML interaction via CCMs (and UCMs) with a focus on the psychological perceptions, behaviors, and qualitative data, answering the call for both qualitative and quantitative research on human-AI/ML interactions [18].

III. METHOD

A. Participants

A power analysis conducted in G*Power [19] revealed that we needed 114 participants to find our effects [repeated-measures ANOVAs with a modest between-subjects effect size ($f = 0.25$), alpha of 0.05, and power set to 0.80] for each study. Participants were randomly assigned to one of the two conditions. We chose a between-subjects platform to avoid problems of task learning influencing our study condition effects. We used CloudResearch [20] to facilitate data collection with MTurk and restricted participants to those who had completed at least 100 human intelligence tasks (HITs) and had at least a 95% approval rate for previous HITs. Eligible participants had to be 18 years of age or older, reside in the United States, and affirm that they were not color deficiency.

In Study 1, a total of 134 participants were collected: oversampling to account for data loss, attrition, and careless responding. After data cleaning, four cases were removed, leaving a total of

130 participants for subsequent analyses. The final sample was 56.15% male and 76.15% white. The average age was 39.62 ($SD = 10.03$). In Study 2, a total of 164 participants were collected, and after data cleaning, 30 cases were removed, leaving a total of 134 participants. The final sample was 61.94% male and 72.39% white. The average age was 39.83 ($SD = 11.30$). In Study 3, a total of 138 participants were recruited. Six cases were removed, leaving a total of 132 participants. The final sample was 57.58% male and 78.79% white. The average age was 38.54 ($SD = 10.63$). In all three studies, participants were compensated with \$6.00 plus any additional bonus earned within the task (see in the following).

B. Task

In all three studies, participants completed two training sessions and ten task pages (i.e., rounds) of an image-binning monitoring task. On each page, participants had 15 s to review 15 images (arranged in a 3×5 matrix) that were classified as cats or dogs by either a UCM or a CCM. There were between three to five images on each page that were incorrectly classified. This range was chosen as we wanted both models to be accurate enough for reliance, so as to determine if the effects were due to the model and not the accuracy of the model. We determined the time of 15 s through pilot testing to allow time for participants to scan most of the images, but not enough to do the entire task themselves. This was because the task is easy for a human, and no condition differences would be present without time constraints. Participants were incentivized to change each incorrect image by starting with a bonus amount of \$10.00 and penalized by \$0.25 per image that was incorrectly classified at the end of the trial. In Studies 2 and 3, participants were informed that the algorithm was trained on images of cats and dogs, but other animals were included in the task round images. Therefore, in Studies 2 and 3, images consisted of images of cats, dogs, and other animals. We only used images of mammals in the other category to avoid drawing attention to the “other” category.

C. Machine Learning Models

The CCM and UCM algorithms used in all three studies were based on the ResNet50 architecture [21] and were implemented using the PyTorch library [22]. Models were trained on 17 000 images of cats and dogs published by Microsoft Research [23]; a full description of the model is available in [24]. We fine-tuned each model, starting from the pretrained ImageNet weights, to adapt them to our specific task. The CCM was trained with outlier exposure and the Objectosphere loss [25]. Meaning, the CCM was exposed to representative out-of-distribution samples during training to improve its confidence estimates. The UCM was trained with the softmax loss function, a standard loss function often used in image classification [e.g., 4, 21], where a class label is assigned to each input. The image data were chosen to ensure easy processing by human. We chose animals as the task would be easy for humans to classify. In other words, we did not want more complex images, such as cancer screening, because participants would need specialized skills, and even with those skills, classification may be inherently difficult.

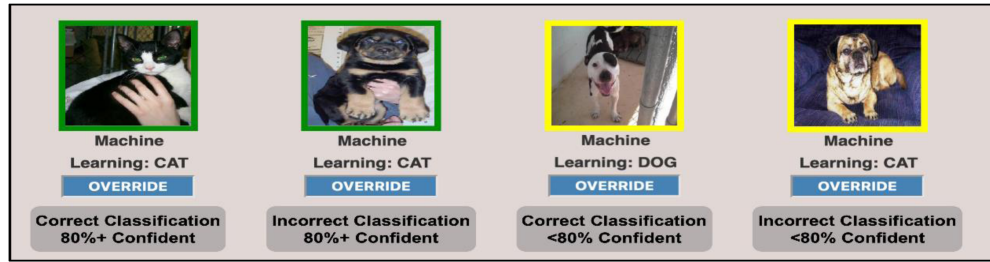


Fig. 1. Four types of images participants encountered during the task, which varied in classification accuracy and confidence.

D. Manipulations

Images were classified by either a UCM or a CCM. Participants were randomly assigned to one of these conditions. Confidence estimates of the models were illustrated with a color border surrounding the picture. High confidence classifications (80% or higher) were bordered in green, and low confidence classifications (below 80%) were bordered in yellow. We dichotomize the confidence level as previous research has demonstrated no differences in trust when reliability is above 80% and marginal differences below 80% [26]. In addition, proper display design of confidence intervals was noted in previous research [27], such that confidence estimates can increase workload when displayed numerically.

In Study 1, for the CCM condition, the algorithm had low confidence in the classification decision when the image was out of distribution. In the UCM condition, the algorithm had consistently high confidence in the classification decision, even if the image was out of distribution. All cats and dogs were classified correctly. All images were the same across conditions. In Studies 2 and 3, for the CCM condition, the algorithm had low confidence in the classification decision when the image was out of distribution (i.e., not a cat or dog). In the UCM condition, the algorithm had consistently high confidence in the classification decision, regardless of the image type. Participants were shown four types of images:

- 1) correct classification and high confidence decision;
- 2) incorrect classification and high confidence decision;
- 3) correct classification and low confidence decision;
- 4) incorrect classification and low confidence decision (see Fig. 1).

The displayed classification decisions as well as the confidence estimates displayed around the images (i.e., green and yellow borders) were actual classification decisions and confidence estimates from a traditional UCM and a CCM.

E. Measures

1) *Task Performance*: We assessed task performance with two metrics: 1) the number of correctly changed images on each page and 2) decision time, measured as the time from the start of the trial to the time of the classification change for each incorrectly classified image.

2) *Perceived Trustworthiness*: We implemented the 15-item system trustworthiness scale (STS) [29] to assess perceived trustworthiness of the models before and after the task. The scale is comprised of three subscales: performance, purpose,



Fig. 2. Example image of round layout and performance information.

and process, with five items per factor. Participants responded to the STS using a 5-point response scale (1 = Strongly Disagree to 5 = Strongly Agree).

3) *Reliance Intentions*: We used the 10-item RI scale [8] to evaluate participants willingness to rely on the models prior to the task. We modified the items so that “ML algorithm” was the referent. The RI scale was assessed on a 7-point response scale (1 = Strongly Disagree to 7 = Strongly Agree).

F. Procedure

The procedure was consistent across all three studies except for key post-task differences, described as follows. Participants completed the entire study online. After initial consent, participants were randomly assigned to one of the two algorithm conditions and read a description of the ML image-binning task. Participants were given an explanation regarding their role in the task, went through several pages of training slides, and then completed a three-item quiz about the task as a knowledge check. Participants who passed the knowledge check then completed two practice rounds and answered the pretask STS and RI measures. After the pretask measures, participants completed the ten experimental rounds of the task. After each round, participants were provided with information regarding their performance during the round, including the number of incorrectly binned images (see Fig. 2), their loss amount, and their current bonus amount.

TABLE I
MEANS, STANDARD DEVIATIONS, AND SCALE RELIABILITIES AT PRETASK AND POST-TASK FOR ALL THREE STUDIES

	Performance			Purpose			Process			Reliance Intentions		
	<i>M</i>	<i>SD</i>	α	<i>M</i>	<i>SD</i>	α	<i>M</i>	<i>SD</i>	α	<i>M</i>	<i>SD</i>	α
Study 1	3.33/3.70	0.83/0.82	0.92/0.93	3.87/4.02	0.61/0.64	0.80/0.89	3.25/3.25	0.93/0.93	0.86/0.89	3.50	1.25	0.93
Study 2	3.13/3.54	0.99/0.96	0.94/0.93	3.70/3.90	0.84/0.72	0.87/0.86	3.26/3.37	0.98/0.95	0.89/0.89	3.32	1.38	0.94
Study 3	3.25/3.51	0.85/0.85	0.89/0.90	3.84/4.07	0.59/0.54	0.80/0.84	3.22/3.21	0.83/0.89	0.84/0.88	3.50	1.29	0.93

Note. Pre-task/post-task.

In Studies 1 and 2, once all ten rounds were completed, participants responded to the post-task STS measure. In Study 3, participants responded to the post-task STS and responded to an open-ended response question, “What strategy or strategies did you use to monitor the ML algorithm?” In all three studies, the participants answered demographic items and were then informed of their bonus amount and compensated via MTurk.

G. Data Cleaning

We utilized several data cleaning checks and metrics to ensure data quality. Upon entering the HIT, we used a CAPTCHA to ensure a human participant and not an automated tool was taking the survey. In addition, participants had to correctly answer all three items on the training quiz to access the rest of the study. We checked for careless responding by assessing page time on surveys, as suggested by Bowling et al. [28] and including instruction items embedded within some surveys (“For this item, select Disagree”) [30]. In addition, we removed participants who did not complete all ten experimental rounds and/or those who did not change at least 10% of the incorrectly labeled images.

IV. RESULTS AND DISCUSSION

All analyses were conducted in the R (version 4.2.3) environment. We used the *lme4* [31] and *emmeans* [32] packages to estimate the models and the marginal means from the models, respectively. The *effectsize* package [33] was used to estimate the variance accounted for in all models. Linear mixed-effects models were performed to analyze the data for each dependent variable, unless another analysis is specified, with condition and round/time as the independent variables. For all analyses, “condition” is the between-subjects manipulation of being paired with a UCM or CCM, “round” is the experimental round where participants viewed the stimuli (ten rounds total), and “time” is the pre- and post-task assessments, which are denoted as first and second, respectively. See Table I for study descriptives and reliabilities. All estimated marginal means and standard errors, as well as pairwise comparisons, are available from the first author.

A. Study 1—No Out-of-Distribution Data

1) *Number of Correctly Changed Images*: There was a significant main effect of round [$F(9, 1152) = 101.44, p < 0.001, \eta_p^2 = 0.44$] and condition [$F(1, 128) = 4.59, p = 0.034, \eta_p^2 = 0.03$] on the number of correctly changed images. The interaction effect was not significant. Results indicated that the number of correctly changed images was significantly higher in the CCM condition than in the UCM condition. Similarly,

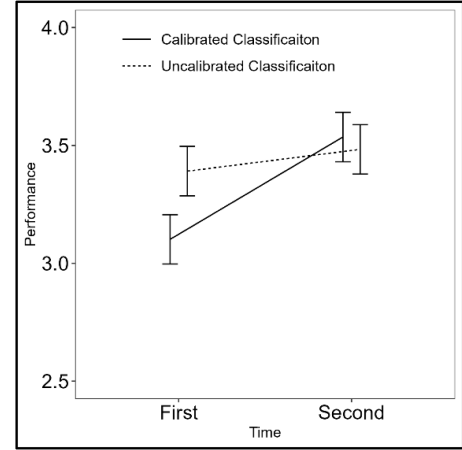


Fig. 3. Performance perceptions over time by condition between CCMs and UCMs.

performance at each time point was significantly different from each other for most observations. Hypothesis 1 was supported.

2) *Decision Time*: We observed a significant main effect of round [$F(9, 4402) = 40.82, p < 0.001, \eta_p^2 = 0.08$]. However, the effects of the condition and the interaction were not significant. Hypothesis 2 was not supported. Decision time was the lowest for rounds 3, 4, 8, and 9 (three incorrect images for each of these rounds), and highest for rounds 2 (five incorrect images), 6 (three incorrect images), and 10 (five incorrect images).

3) *Trustworthiness*: First, we analyzed performance perceptions. We observed a significant main effect of time [$F(1, 128) = 13.41, p < 0.001, \eta_p^2 = 0.09$]. The main effect of the condition was not significant. The main effect of time was qualified by a condition-by-time interaction [$F(1, 128) = 5.65, p = 0.019, \eta_p^2 = 0.04$]. As illustrated in Fig. 3, participants in the CCM condition had lower initial performance perceptions than the UCM condition. However, after interacting with the output of the algorithm for ten rounds, there were no differences between the UCM and CCM conditions. Hypothesis 3 was not supported. Next, we performed analyses on purpose perceptions. We observed a significant main effect for time [$F(1, 128) = 23.74, p < 0.001, \eta_p^2 = 0.16$] and an interaction between time and condition [$F(1, 128) = 7.73, p = 0.006, \eta_p^2 = 0.06$]. As illustrated in Fig. 4, there were marginal differences between CCM and UCM conditions at both pretask and post-task assessments. However, whereas participants’ purpose perceptions in the UCM condition were slightly higher during the pretask assessment, purpose perceptions in the CCM condition were slightly higher during the post-task assessment. The main effect of the condition was

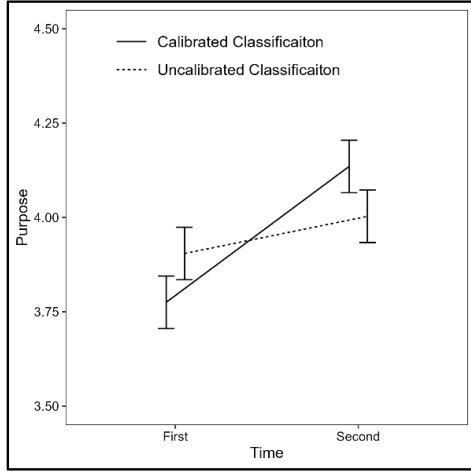


Fig. 4. Purpose perceptions over time between CCMs and UCMs.

TABLE II
HIERARCHICAL REGRESSIONS PREDICTING RI

Variable	B	SE	95% CI for B		β	R^2	ΔR^2
			LL	UL			
<i>S1. Step 1</i>							
Constant	3.28**	0.16	2.97	3.59		.03	
Condition	0.43 [†]	0.22	-0.10	0.87	0.17		
<i>S1. Step 2</i>							
Constant	0.51	0.72	-0.92	1.95		.35	.38**
Condition	0.48**	0.18	0.12	0.84	0.19**		
Performance	0.79**	0.14	0.51	1.03	0.52**		
Purpose	-0.30	0.22	-0.72	1.06	-0.12		
Process	0.37**	0.11	0.15	0.59	0.25**		
<i>S2. Step 1</i>							
Constant	3.46**	0.16	3.14	3.77		.00	
Condition	0.08	0.22	-0.35	0.51	0.03		
<i>S2. Step 2</i>							
Constant	-0.25	0.56	-1.36	0.85		.41	.41**
Condition	0.17	0.18	-0.18	0.52	0.07		
Performance	0.79**	0.15	0.49	1.09	0.52**		
Purpose	-0.07	0.19	-0.45	0.30	-0.04		
Process	0.32**	0.11	0.10	0.54	0.24**		
<i>S3. Step 1</i>							
Constant	3.97**	0.17	3.64	4.30		.01	
Condition	-0.3	0.24	-0.77	0.16	-0.11		
<i>S3. Step 2</i>							
Constant	-0.09	0.39	-0.86	0.68		.67	.66**
Condition	-0.16	0.14	-0.44	0.11	-0.06		
Performance	1.07**	0.10	0.87	1.27	0.76**		
Purpose	-0.20	0.14	-0.49	0.08	-0.11		
Process	0.30**	0.09	0.11	0.48	0.21**		

Note. CI = confidence interval; LL = lower limit; UL = upper limit. ** = $p < .01$, [†] = $p < .10$. S1 = Study 1, S2 = Study 2, and S3 = Study 3.

not significant. Finally, we explored process perceptions. No main effects or interaction effects were significant.

4) *Reliance Intentions*: We conducted a post-hoc hierarchical multiple regression analysis to determine if trustworthiness perceptions (i.e., performance, purpose, and process) provided additional information in predicting RI, after controlling for condition. In step 1, condition was a marginally significant predictor of RI [$F(1, 128) = 3.73$, $p = 0.056$, $R^2 = 0.03$]. In step 2, trustworthiness perceptions explained significant additional variance in RI [$F(4, 125) = 19.38$, $p < 0.001$, $R^2 = 0.38$], after controlling for condition. Results are given in Table II. In the final regression, condition ($\beta = 0.48$, $p = 0.009$), performance ($\beta = 0.79$, $p < 0.001$), and process perceptions ($\beta = 0.37$, $p = 0.001$) were unique significant predictors of RI; however, purpose perceptions were not a significant predictor.

B. Study 2—Out-of-Distribution Data Included

1) *Number of Correctly Changed Images*: There was a significant main effect of round on task performance [$F(9, 1188) = 206.70$, $p < 0.001$, $\eta_p^2 = 0.61$]. The main effect of the condition was marginally significant. Furthermore, the condition by round interaction was significant [$F(9, 1188) = 6.77$, $p < 0.001$, $\eta_p^2 = 0.05$]. We performed pairwise comparisons of each condition across rounds. Results indicated a significant difference at round 2 [$t(728) = 6.14$, $p < 0.001$] and round 10 [$t(728) = 3.37$, $p < 0.001$] such that participants in the CCM condition performed better than participants in the UCM condition. It is important to note, rounds 2 and 10 had the greatest number of incorrectly classified images to be changed. Hypothesis 1 was partially supported, such that participants did perform significantly better in the CCM versus UCM condition, although this difference was only seen in rounds 2 and 10.

2) *Decision Time*: We observed a significant main effect for condition [$F(1, 129.5) = 19.19$, $p < 0.001$, $\eta_p^2 = 0.13$] and round [$F(9, 4195) = 59.83$, $p < 0.001$, $\eta_p^2 = 0.11$]. Pairwise comparisons across conditions revealed participants in the CCM condition took less time to respond [$t(136) = -4.38$, $p < 0.001$], supporting Hypothesis 2. Decision time was the lowest for rounds 3, 4, 8, and 9 (three incorrect images for each of these rounds), and highest for rounds 2 (five incorrect images), 6 (three incorrect images), and 10 (five incorrect images). However, the condition-by-round interaction was not significant.

3) *Trustworthiness*: A significant main effect of time was observed for all three models [performance: $F(1, 132) = 37.19$, $p < 0.001$, $\eta_p^2 = 0.22$; purpose: $F(1, 132) = 13.83$, $p < 0.001$, $\eta_p^2 = 0.09$; process: $F(1, 132) = 8.58$, $p = 0.004$, $\eta_p^2 = 0.06$], such that participants' trustworthiness perceptions increased over time from pre-task to post-task. No other main effects or interaction effects were significant. Hypothesis 3 was not supported; participants did not perceive the performance of the CCM to be higher than the UCM. In addition, participants did not rate the CCM significantly different on purpose or process perceptions.

4) *Reliance Intentions*: We conducted a post-hoc hierarchical model to determine if trustworthiness perceptions provided additional information in predicting RI, after controlling for condition. In step 1, condition was not a significant predictor of RI [$F(1, 132) = 0.14$, $p = 0.705$, $R^2 < 0.01$]. In step 2, trustworthiness perceptions explained significant additional variance in RI [$F(4, 129) = 22.07$, $p < 0.001$, $R^2 = 0.41$] after controlling for the condition. Results are illustrated in Table II. In the final regression, performance ($\beta = 0.52$, $p < 0.001$) and process perceptions ($\beta = 0.24$, $p = 0.004$) were unique significant predictors of RI; however, purpose perceptions were not a significant predictor.

C. Study 3—Qualitative Study

1) *Number of Correctly Changed Images*: There was a significant main effect of round [$F(9, 1170) = 319.80$, $p < 0.001$, $\eta_p^2 = 0.55$]. Participants' performance at each time point was significantly different from each other for most observations. The main effect of the condition and the interaction term of

round by condition were not significant. Hypothesis 1 was not supported; participants did not perform significantly better when monitoring the CCM versus the UCM.

2) *Decision Time*: We observed a significant main effect of condition [$F(1, 117.90) = 5.74, p = 0.018, \eta_p^2 = 0.05$]. Pairwise comparisons across conditions revealed participants in the CCM condition took less time to respond than the UCM condition [$t(132) = -2.40, p = 0.018$]. We also observed a significant main effect of round [$F(9, 4019.50) = 44.49, p < 0.001, \eta_p^2 = 0.09$]. Decision time was the lowest for rounds 3, 4, 8, and 9 (three incorrect images for each of these rounds), and highest for rounds 2 (five incorrect images), 6 (three incorrect images), and 10 (five incorrect images). However, the condition by round interaction was not significant. Hypothesis 2 was supported.

3) *Trustworthiness*: First, we analyzed performance perceptions. We observed a significant main effect for time [$F(1, 130) = 23.61, p < 0.001, \eta_p^2 = 0.15$]. The pairwise comparisons demonstrate participants' performance perceptions increased over time from pretask to post-task [$t(130) = -4.86, p < 0.001$]. The main effect of the condition and the interaction were not significant. Hypothesis 3 was not supported. Next, we performed analyses on purpose perceptions. We observed significant main effects for time [$F(1, 130) = 8.85, p = 0.003, \eta_p^2 = 0.06$]. Participants' purpose perceptions increased significantly over time from pretask to post-task [$t(130) = -2.98, p = 0.004$]. The main effect of the condition and the interaction effect were not significant. Finally, we explored process perceptions. The main effect of time, condition, and the interaction were all not significant.

4) *Reliance Intentions*: We conducted a post-hoc hierarchical multiple regression analysis to determine if trustworthiness perceptions provided additional information in predicting RI, after controlling for condition. In step 1, condition was not a significant predictor of RI [$F(1, 130) = 1.66, p = 0.198, R^2 = 0.01$]. In step 2, trustworthiness perceptions explained significant additional variance in RI [$F(4, 127) = 63.93, p < 0.001, R^2 = 0.66$], after controlling for condition. Results are given in Table II. In the final regression, performance ($\beta = 1.07, p < 0.001$) and process perceptions ($\beta = 0.30, p = 0.002$) were unique significant predictors of RI; however, purpose perceptions were not a significant predictor.

5) *Qualitative Analysis*: We asked participants, "What strategy or strategies did you use to monitor the ML algorithm?" to supplement our quantitative analyses and better understand the differences in how people approached the task. Two raters conducted a thematic analysis of participants' text responses. We denoted the broader themes identified in our analyses with italics and specific codes with quotations (see Table III). Frequencies of identified themes and codes within responses were calculated for each condition, and a 30% absolute difference cutoff was applied, using the UCM condition as a reference. We decided to use this percentage benchmark, as it provided a more conservative and standardized comparison to better interpret condition differences.

We found that some aspects of participants' strategy were consistent across both conditions. For example, many participants mentioned using some form of a visual search strategy

TABLE III
QUALITATIVE ANALYSIS RESULTS

Theme	Sub-Theme	Code
Focus	Labels*	label/name*
	Image*	cat/dog image, non-cat/dog image*, picture/image*
	Color*	yellow*, green, yellow $\rightarrow$ green*, green $\rightarrow$ yellow*
	Confidence*	confidence border (high/low)*
Task	Visual Search	scan, skim, pattern
	Strategy Change	learning, strategy change, multiple strategies
	Time	quickly*, time pressure
Algorithm Evaluation	Accuracy	algorithm accuracy (high/low)
	Trust	trust (high/low)*
	Algorithm Use*	did use, did not use*

Note. *denotes condition differences in user evaluations of UCMs and CCMs utilizing the 30% difference threshold.

involving "scanning" or "skimming" across different aspects of the task display. Some described using a specific visual scan "pattern" (e.g., "left-to-right" or "row-by-row"). Many participants also indicated some form of strategy change when completing the task. This included an indication of "learning" something new, adding an "additional strategy," or adopting a "different strategy." For example, one CCM user wrote, "At first I tried to give each image a good look before moving to the next. This was taking too much time, so I decided to concentrate only on the yellow-boxed images and monitor how much better my performance was..." Participants described both UCMs and CCMs as having "high accuracy" and/or "low accuracy."

We also identified key differences in participants' strategies between conditions. These differences primarily involved participants' main focus during their strategy process. UCM users described focusing on the actual animal image 77% more often than CCM users. Many participants in the UCM condition specifically referenced the out-of-distribution images that were "not cats or dogs" (48.15% more). For instance, participants recounted, "I looked for an animal that wasn't dog or cat..." and "I quickly noticed it was doing a good job at cat or dogs so only looked for photos that were not a cat or dog..." UCM users also mentioned utilizing the task image's label/name 57.14% more often than CCM users. For instance, one participant described, "...tried to look at every picture and how it was labeled." Others recalled a mixed-approach utilizing both images and labels such as, "I just had to glance from photo to label." These trends indicated less user reliance on the algorithm when completing the task as their responses were more focused on the aspects of the task display that were non-related to the algorithm(s). In fact, when describing algorithm use, participants in the UCM condition directly stated they "did not utilize the algorithm," 150% more often than those in the CCM condition. For example, one participant stated, "i just ignored it, it wasn't accurate enough in general." In contrast, participants working with CCMs mentioned color in their descriptions 68% more frequently than those working with UCMs. Many participants in this condition stated they specifically focused on the "yellow" border surrounding the images, 58.33% more often. For instance, one participant explained, "I first went to any animals that were given a yellow border, and corrected them immediately." Furthermore,

other CCM users described utilizing both colors in their strategy, including focusing on “yellow borders to green borders” (100% more than UCMs) and “green borders to yellow borders” (100% more than UCMs). Similarly, participants in the CCM condition mentioned focusing on the algorithm’s confidence when completing the task, 60.87% more than those in the UCM condition. These participants specifically reference the “high confidence” algorithm decisions (100% more than UCMs) and “low-confidence” algorithm decisions (70% more than UCMs). For example, one participant said, “I focused on photos w[h]ere the algorithm was less than 80% confident, then I quickly scanned those w[h]ere it was above 80%.” Similarly, another participant primarily focused on “...the low confidence level the machine predicted.” These descriptions of a color-focused strategy indicated greater utilization of and reliance on the CCM algorithm.

We also identified that CCM users had more prevalent descriptions of trust toward the algorithm. The trust theme in this analysis encompassed various descriptions involving general trust attitudes, RI, and reliance behaviors toward the UCM/CCM. This included 40% more descriptions of “high trust” descriptions and 100% fewer descriptions of “low trust” descriptions toward CCMs. For instance, one CCM user stated, “After a few rounds, I started trusting the algorithm more and more and only focused on checking the yellow images.” Another stated, “I worked on the low confidence level the machine predicted since I can rely on the prediction of the high confidence level.” Furthermore, when describing general characteristics of task completion, we found more CCM descriptions regarding time and users completing the task “quickly,” 35.14% more than in the UCM condition.

V. GENERAL DISCUSSION

Across three studies, we explored differences in task performance and psychological perceptions between traditional UCM and CCM models in an image-binning monitoring task. As noted in previous reviews [1] and work [4], XAI and CCMs are being developed to help users better understand the decision processes of AI/ML models. Results from the present studies partially support the hypothesis that performance transparency, as implemented with CCMs, increases actual performance on a task. However, the current study also illustrates several issues with attempting to assess user perceptions of AI/ML models. In general, the findings support the theoretical models put forth by Alarcon and Willis [3] and Lee and See [7], such that adding transparency to an aspect of trustworthiness can influence the trust process.

A. Task Performance

In two out of three studies, when differences in task performance were observed, participants in the CCM condition performed better than participants in the UCM condition. This finding aligns with previous research that has found transparency increases user performance on a task [12], [34]. In the more difficult rounds (i.e., five images to correct), task performance differed between conditions. Research has suggested that two contributing factors to automation use and reliance are workload

and task difficulty, for which automation is utilized more often to maintain pace with task demands (e.g., [35]). When the task was more difficult, task performance increased with the help of the CCM.

In general, decision time was significantly faster for the CCM condition in Studies 2 and 3, demonstrating the usability of the transparent model for quicker decision-making capabilities. These results provide insight into how participants relied upon the CCM by accepting its decisions instead of taking the extra time to scan each image, which could be considered a behavioral outcome of trust (i.e., reliance behaviors [7]). Indeed, the qualitative analyses demonstrated that participants using the CCM focused on low confidence images first then other images. In contrast, participants in the CCM condition scanned all the images as they did not trust the system. This can be further described as compliance, in that participants allowed the model to inform their decisions and accepted the model’s highly confident decisions [17], [36]. Results demonstrated that participants had higher reliance and compliance with the more transparent model. Interestingly, decision time was not significant for Study 1. A potential explanation for this is that there were only images of cats and dogs being flagged with low confidence. Therefore, even with the more accurate confidence estimates, participants still needed to take more time to check the images because they were similar to the correctly binned images. When viewing this through the lens of signal detection theory, this could correspond to decision uncertainty on whether the flagged images were false alarms or correct rejections because the images were all cats or dogs. Users are less likely to comply with the alerts and take the extra time to cross check the decisions [37]. Therefore, even with the additional confidence information, decision time would be comparable to the less transparent condition.

B. Trustworthiness and Reliance Intentions

In Studies 2 and 3, condition did not influence trustworthiness perceptions. This finding is interesting considering that task performance and decision time were improved by using the CCM over the UCM. In Study 1, when participants only viewed in-distribution data, performance perception differences were found. Participants initially perceived the UCM as having increased performance and purpose after training. This demonstrated that participants perceived the lack of confidence by the CCM as a trustworthiness hindrance. However, after interacting with the models across the task, participants perceived no performance differences between the two models, which contrasts with previous theoretical classifications of CCMs [3]. Participants perceived an incorrect classification as the same, regardless of model confidence. CCMs were developed to indicate low confidence in out-of-distribution images. In addition, purpose perceptions were higher in the CCM condition than in the UCM condition in the posttask assessment. This may be because model performance is explicitly linked to purpose in many human-machine teaming tasks [3]. Interestingly, there were no effects of condition or round on process perceptions in either of the conditions. This finding has multiple implications for research. Transparency of the model’s process was not facilitated in the

current study, such that an underlying explanation of how the decision was made by each model was not presented to the user. As Alarcon and Willis [3] noted, XAI is linked to the underlying processes of how the decision is made. The fact that process perceptions remained constant across the study illustrates that the psychological process perception is different from the performance and purpose perceptions, as it was not transparent. This more nuanced approach to facilitating trustworthiness, besides just performance, has been called for by researchers over the last few years [29], [38], [39].

Finally, RI were examined post-hoc to investigate whether perceptions of trustworthiness added incremental variance to intentions to rely on the model after accounting for condition. Specifically, in all three studies, performance and process were unique predictors of RI, as theorized by Lee and See [7]. As noted by previous research works [3], [40], performance is a key aspect of trust in machines and algorithms. If the system does not perform the task well, reliance on the system will decrease. Process perceptions were also a unique predictor in RI, indicating users who felt more confident in understanding how the models came to their decisions were more likely to rely on them. Theory posits that providing more accurate confidence estimates increases the performance transparency of the system, which may have also led the user to infer an understanding of the underlying processes that guide the system's decision making. However, purpose was not a significant predictor in any of the three studies, demonstrating that purpose did not explain any additional variance in RI after accounting for the ML type and perceptions of performance and process. Prior research has suggested that performance and purpose are interrelated [29], so a possible explanation for the nonsignificance of purpose is that any additional variance explained by purpose was suppressed by the variance explained by performance.

C. Qualitative Analysis

Many of the themes found involved general characteristics regarding the nature of task completion, such as visual scanning behavior and strategy adjustments. We corroborated our quantitative findings in Studies 2 and 3 regarding a lack of differences for performance perceptions between UCM and CCM users. The qualitative analysis allowed us to better differentiate other aspects of users' perceptions, evaluations, and strategies when interacting with the algorithms. Overall, there were generally more positive descriptions ascribed to CCMs than UCMs. We found that users utilized different strategies when completing the task; UCM users described a more manual check of each image, and CCM users relied more on the algorithms' colored confidence displays, indicating higher levels of trust toward CCMs. We identified other potential benefits CCM users experienced, such as quicker and/or more efficient task completion. This aligns with our lower decision time findings from Study 2 and Study 3 associated with CCM users. Although these perception differences were not identified with some of our inferential statistics, additional research is warranted to further explore these interactions, especially given the higher prevalence of trust descriptions from CCM users. Participants' evaluations and actual behavioral outcomes, based on qualitative descriptions,

indicate possible variations between conditions. In other aspects of user interaction, we found qualitative evidence validating our quantitative analyses, such as similar trustworthiness perceptions and faster CCM decision times. Both findings highlight the utility of qualitative methods in offering additional insight into the differences in human-machine interaction between UCMs and CCMs, complementing quantitative approaches.

D. Limitations and Future Directions

We note several limitations of the current study. First, we utilized novices for the current task. Experienced practitioners of AI/ML may respond differently to the algorithms' confidence levels. Second, and related to the first point, we utilized the green and yellow color-coding system for high and low confidence levels, respectively. We chose this dichotomy of the confidence intervals to facilitate quicker processing of the information than numerical data. It remains to be seen whether there are differences between numerical confidence estimates and color-coded estimates. Practitioners of AI/ML may appreciate more nuance in the output of the model, such as the exact confidence estimates. Third, our results demonstrated that there were fewer significant differences in rounds of lower task difficulty (fewer corrections to be made). We set the DNNs' accuracy to 80%, per previous research [24], [40]; however, the task was easy enough for the participants to perform all the image classifications during rounds with fewer changes needed to be made. Therefore, future work would benefit from implementing the experiment in a more difficult task. For example, classification of planes as military or civilian or cancer screening may offer platforms of AI/ML that are not as easily discernible. However, we were interested in the effects of CCMs over UCMs and did not want to conflate image content area with perceptions of the models. Another avenue could be to decrease the performance of the models. Both models were highly accurate. It may be that CCMs are more important when the models are less accurate. Finally, we utilized a between-subjects design across the three studies, where participants only interacted with one algorithm. Given their limited experience with AI/ML and introduction to a novel classification task, participants lacked a frame of reference to compare their experiences with both UCMs and CCMs, which may have contributed to some of the similar perceptions and evaluations found between conditions in this study. However, for these studies, we wanted to ensure that differences between conditions were due to experimental stimuli and not from learning the classification task, which could have complicated results. Future studies would benefit from implementing a within-subjects design, allowing participants the opportunity for interaction and comparison of both algorithm types.

E. Conclusion

We aimed to investigate how AI/ML models with transparency of confidence estimates impacted task performance and trustworthiness perceptions in an image classification task across three studies. Results demonstrated that when participants interacted with classifications that only considered in-distribution data, the increased transparency of CCMs led to increased task performance in Study 1, and in Study 2, when task difficulty

was increased. Performance and purpose perceptions differed depending on whether it was assessed pretask or posttask, but only for one study. When the CCMs included out-of-distribution data, we found differential effects for task performance when task difficulty was heightened. Decision time was also lower for CCMs compared to traditional UCMs. Although there was no consistent pattern of significant differences between conditions in our quantitative results for trustworthiness perceptions, an examination of the qualitative responses indicated that participants found the CCMs to be more trustworthy, and they utilized different strategies (e.g., focusing on the low confidence borders) when compared to the UCMs. Overall, the current research demonstrated partial support for a priori hypotheses and emphasizes the importance of ML/AI decisions to include additional information with more accurate confidence estimates to increase trust and task performance.

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